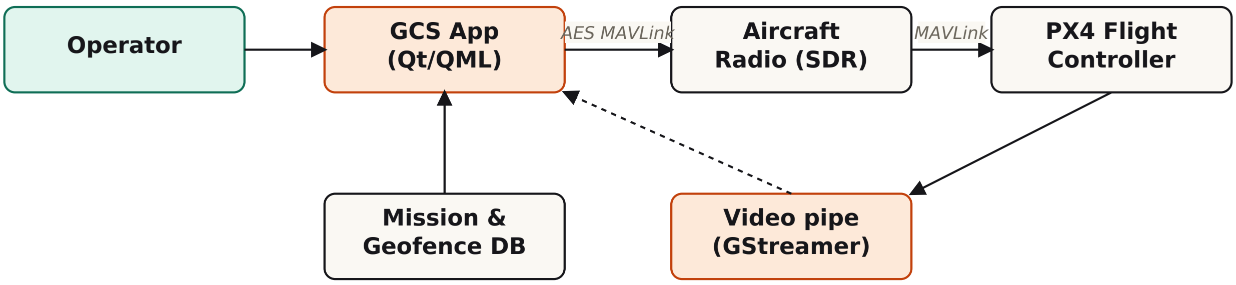
GROUP A . MULTI-DISCIPLINARY · PROJECT 01 OF 14

Military-Grade Ground Control Station

GCS · UAV · defence software

A modernised, hardened ground control station built from the ground up for Indian defence operations. Multi-vehicle ops, AES-256-encrypted MAVLink, mission planning with no-fly zones and offline tile maps, a 1080p video pipeline, and a rugged enclosure that survives field deployment.

ARCHITECTURE



WHAT YOU'LL BUILD

- Qt/QML GCS desktop application supporting 5+ simultaneous vehicles
- Encrypted MAVLink transport (AES-256 + PGP key exchange)
- Mission planner with no-fly zones, geofences, offline tile cache
- Live video pipeline (GStreamer) with HUD overlay
- Rugged enclosure housing SBC, display, RF interface, battery
- Field test report: 5+ vehicles, 30+ km link distance

SKILLS YOU'LL GAIN

- Qt/QML application architecture
- MAVLink protocol design
- Practical cryptography (AES, key management)
- GStreamer pipelines
- Rugged hardware integration

TIME

PER WEEK	8–10 hrs
DURATION	12+ months

PREREQUISITES

Distributed across team

OUTCOMES

Working prototype · Defence-grade product

COMMERCIAL OUTLOOK

High. Defence procurement target.

TECH STACK

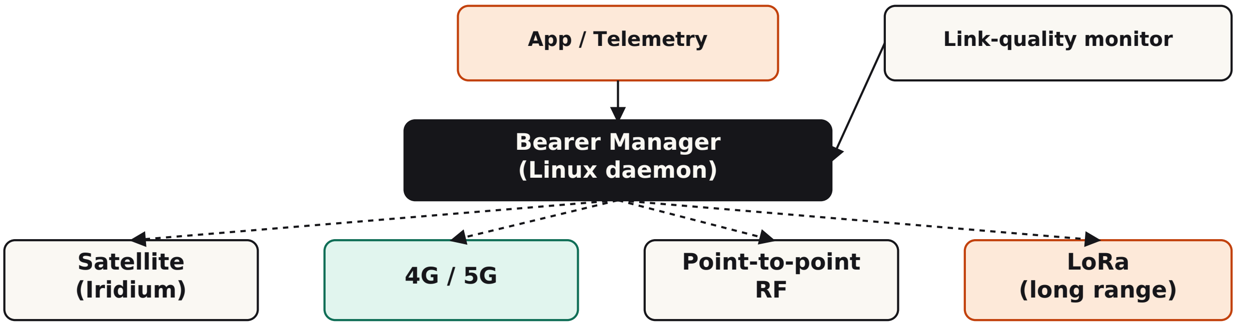
C++/Qt · pymavlink · GStreamer · AES/PGP

Multi-Bearer Communications (Sat + 5G + RF + LoRa)

Hybrid resilient comms · low/high bandwidth

A comms layer that fails over seamlessly between satellite (Iridium), cellular (4G/5G), point-to-point RF, and LoRa based on live link-quality measurement. Handles low-bandwidth telemetry and high-bandwidth video.

ARCHITECTURE



WHAT YOU'LL BUILD

- Bearer-manager daemon routing IP over the best available link
- QoS classifier: video on high-BW only; telemetry on any link
- Live link metrics covering RSSI, latency, jitter, and packet-loss
- Hot failover demo: 4G drop to LoRa with no app-level reconnect
- Field test report: bearer transitions across a 10+ km flight
- Conference-grade paper on the failover algorithm

SKILLS YOU'LL GAIN

- Linux networking internals (netfilter, namespaces)
- Modem management
- Radio propagation and link-budget analysis
- Userspace routing
- Research-paper writing

TIME

PER WEEK	8 hrs
DURATION	12+ months

PREREQUISITES

Linux networking, RF basics, embedded C

OUTCOMES

Research paper · Defence-grade module

COMMERCIAL OUTLOOK

Very high. Defence plus remote and disaster operations.

TECH STACK

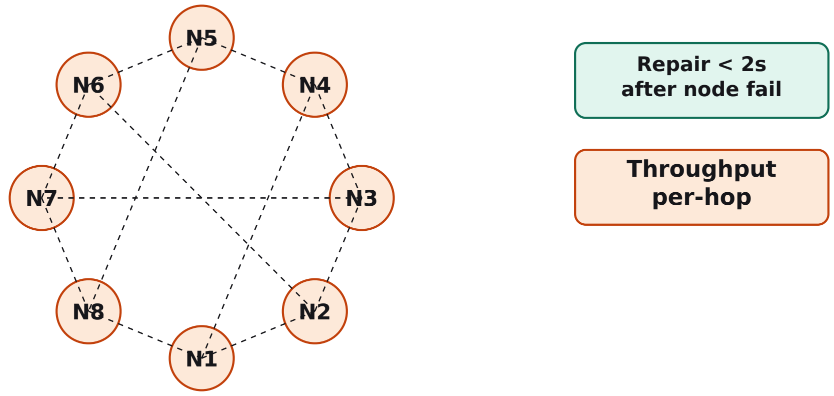
Linux · ModemManager · custom routing daemon · GNU Radio

MANET (Mobile Ad-hoc Network) for Drone Swarms

Mesh networking · routing protocols

A self-forming, self-healing wireless mesh across 8+ nodes. Implement BATMAN-adv or a custom OLSR variant in user space. Characterise throughput, latency, and repair time. Produce a research-grade dataset and analysis.

ARCHITECTURE



WHAT YOU'LL BUILD

- 8-node mesh deployment (ESP32 or Pi-based)
- Routing protocol: BATMAN-adv tuned, or custom OLSR-derived
- Self-test harness: simulate node failures, measure repair time
- Throughput benchmarks at 1, 3, 5 hops under varying load
- Loss-rate dataset under mobility
- Research-grade analysis report

SKILLS YOU'LL GAIN

- Mesh routing protocols (OLSR, BATMAN, AODV)
- 802.11s and ad-hoc mode
- Network measurement methodology
- Embedded Linux at scale

TIME

PER WEEK	6 hrs
DURATION	6–9 months

PREREQUISITES

Linux networking, packet protocols

OUTCOMES

Research paper · Reusable mesh stack

COMMERCIAL OUTLOOK

Medium-high. Defence, agritech, mesh IoT.

TECH STACK

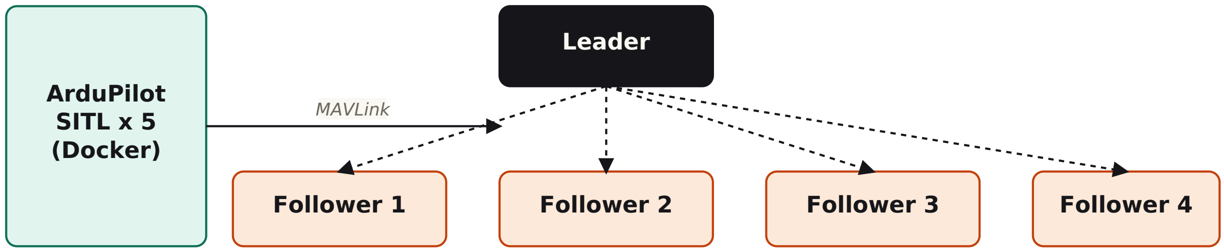
Linux netfilter · BATMAN-adv · Python · Wireshark

Drone Swarm / Multi-Agent System

Multi-agent control · formation flight

A coordinated drone swarm covering formation hold, leader election, and distributed task allocation. Initial development entirely in SITL (zero hardware risk). Optional Phase 2 on Crazyflie-class micro-drones. Same architecture used in production multi-agent UAV ops.

ARCHITECTURE



WHAT YOU'LL BUILD

- 5+ instance ArduPilot SITL orchestration (Docker Compose)
- Formation-hold algorithm: line, triangle, leader-follower
- Distributed leader election under leader failure
- Real-time swarm visualiser (React + WebSockets)
- Metrics dashboard: position drift, formation error
- Optional Phase 2: hardware migration to 5 Crazyflies

SKILLS YOU'LL GAIN

- Multi-agent systems theory
- ROS2 architecture and DDS
- asyncio Python
- ArduPilot SITL internals
- React + WebSockets for live viz

TIME

PER WEEK	7 hrs
DURATION	9+ months

PREREQUISITES

ROS2, Python, basic control theory

OUTCOMES

Research paper · Industry-mentored · Internship-ready

COMMERCIAL OUTLOOK

High. Logistics, ISR, agritech.

TECH STACK

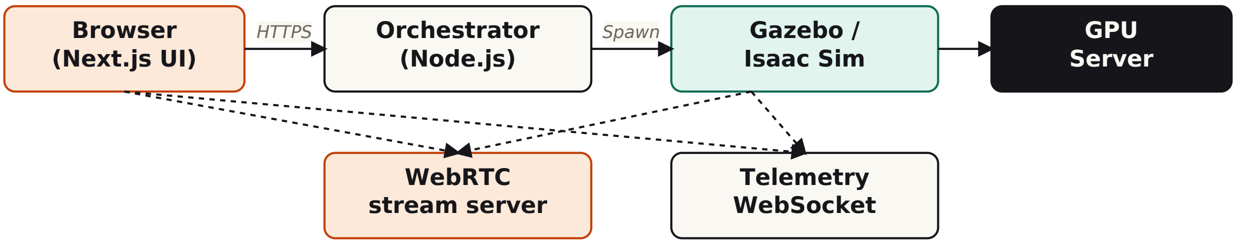
ROS2 · Python · ArduPilot SITL · MAVROS

Web-based SITL World Model (Gazebo / Isaac Sim)

Simulation infrastructure · sim-as-a-service

Browser-accessible Gazebo and Isaac Sim. Students click a button, pick a world, and start running missions in a simulated drone or rover without installing 8 GB of sim software. Becomes the lab's standard onboarding tool.

ARCHITECTURE



WHAT YOU'LL BUILD

- Containerised Gazebo + Isaac Sim that boots on demand
- Worlds library: urban, rural, indoor warehouse, GPS-denied
- WebRTC pipeline: server-rendered viewport at 30 fps
- Telemetry stream over WebSocket to browser; REST for replays
- Session management with scenario picker
- Next.js frontend with live viewer

SKILLS YOU'LL GAIN

- Linux orchestration (Docker, GPU passthrough)
- Gazebo / Isaac Sim authoring
- WebRTC server design
- Next.js full-stack
- GPU resource scheduling

TIME

PER WEEK	7 hrs
DURATION	9+ months

PREREQUISITES

Python, web stack, Linux

OUTCOMES

Internal platform · Open-source release · Skill development

COMMERCIAL OUTLOOK

Medium. Sim-as-a-service for defence and agri startups.

TECH STACK

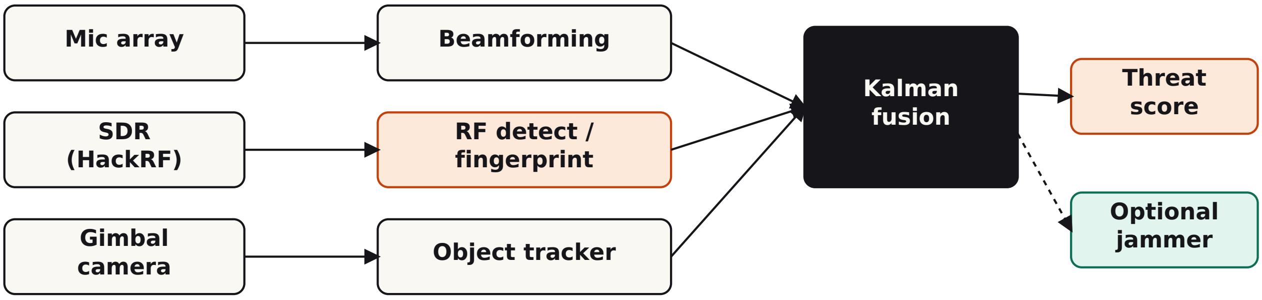
Gazebo Garden · Isaac Sim · WebRTC · Next.js

Anti-Drone Detection System

Counter-UAS · RF + acoustic + optical fusion

A complete counter-UAS detection system fusing RF spectrum analysis, microphone-array beamforming, and optical tracking. Outputs which drones are present, their type, heading, and threat score. Optional soft-kill via RF jamming, with proper legal framing.

ARCHITECTURE



WHAT YOU'LL BUILD

- SDR-based RF detection of drone control links
- Mic-array beamforming for directional acoustic localisation
- Camera + gimbal tracking once a target is acquired
- Sensor fusion: Kalman filter combining all three modalities
- Threat-scoring algorithm (proximity, type, behaviour)
- Optional: RF-jammer prototype in an RF-shielded room

SKILLS YOU'LL GAIN

- SDR signal processing (GNU Radio)
- DSP fundamentals (beamforming, FFTs)
- Sensor fusion (Kalman, particle filters)
- Computer vision with gimbal control

TIME

PER WEEK	8 hrs
DURATION	12+ months

PREREQUISITES

Distributed across team

OUTCOMES

Defence-grade product · Research paper

COMMERCIAL OUTLOOK

Very high. Airports, defence, critical infrastructure.

TECH STACK

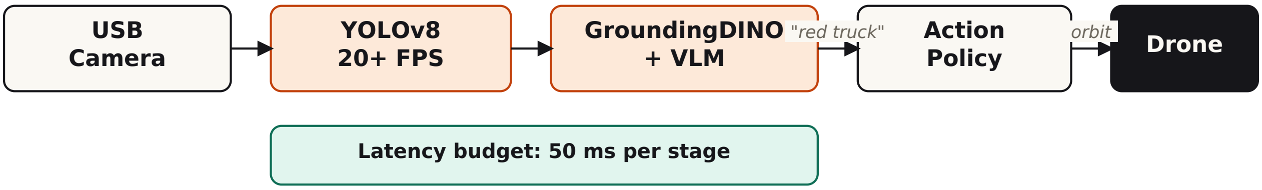
GNU Radio · PyTorch · OpenCV · Kalman filters

Perception Stack with VLA World Model

Edge AI · vision-language-action models

An onboard perception stack for autonomous drones. YOLOv8 for speed combined with vision-language-action models for open-vocabulary understanding ("find the red truck near the warehouse entrance"). Targets low latency on low-cost hardware.

ARCHITECTURE



WHAT YOU'LL BUILD

- Detection pipeline at 20+ FPS on Jetson Orin Nano
- VLM integration: prompt-based open-vocabulary detection
- Action policy: orbit target, follow, return-to-base
- Latency-budget analysis with each stage measured
- Live demo: drone executes a natural-language instruction
- Optional research paper on latency-vs-accuracy

SKILLS YOU'LL GAIN

- PyTorch / ONNX edge deployment
- Quantisation and pruning
- VLM prompting and grounding
- Latency profiling
- ROS2 / message-bus design

TIME

PER WEEK	7 hrs
DURATION	9+ months

PREREQUISITES

PyTorch, CV fundamentals

OUTCOMES

Research paper · Working prototype · Internship-ready

COMMERCIAL OUTLOOK

Very high. Autonomous drones, robotics.

TECH STACK

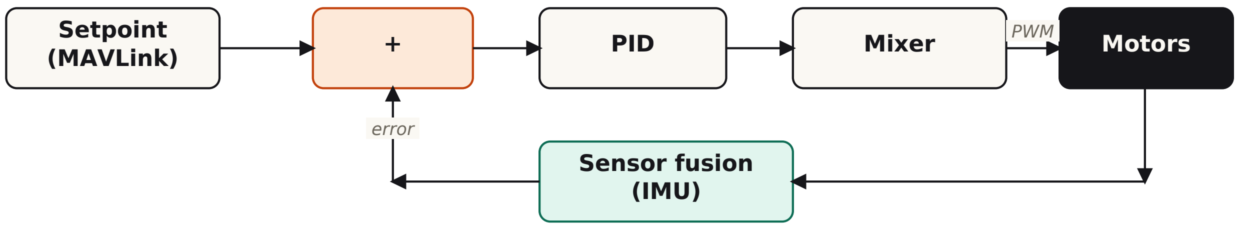
PyTorch · YOLOv8 · GroundingDINO · RT-2 / Octo

Drone FC Firmware Module

Embedded firmware · attitude control

Pick one piece of a flight controller and own it end-to-end: sensor fusion, the PID inner loop, the motor mixer, or the telemetry parser. Your module drops into PX4 or Betaflight via the standard module interface, so you're contributing to real open-source firmware.

ARCHITECTURE



WHAT YOU'LL BUILD

- One firmware module: sensor fusion, PID, mixer, OR telemetry
- Unit tests + hardware-in-the-loop bench validation
- Documentation: math derivation + module API
- Bench-test video showing stable hover with your module live
- Bonus: pull request against upstream PX4 / Betaflight

SKILLS YOU'LL GAIN

- Embedded C in real-time
- RTOS basics (NuttX)
- IMU and sensor-fusion math
- PID tuning
- Hardware-in-the-loop testing

TIME

PER WEEK	5 hrs
DURATION	3–6 months

PREREQUISITES

Embedded C, basic control

OUTCOMES

Working module · Skill development

COMMERCIAL OUTLOOK

Medium. Highly valued embedded skill.

TECH STACK

C · NuttX / Betaflight · MAVLink

GPS / GNSS Driver Module

GNSS · embedded driver

A clean GNSS driver for u-blox modules. NEO-M9N for standard work, ZED-F9P for centimetre-level RTK. The driver parses the UBX binary protocol natively (not just NMEA), supports multi-constellation, and if F9P, implements RTK base/rover.

WHAT YOU'LL BUILD

- UBX protocol parser with full navigation message coverage
- Multi-constellation configuration and selection
- Optional RTK base/rover pair (F9P only)
- Driver library packaged for Linux (Pi) and ESP32
- Documentation and usage examples
- Open-source release on GitHub

SKILLS YOU'LL GAIN

- Binary protocol parsing
- GNSS fundamentals
- RTK theory and RTCM corrections
- Library packaging
- Open-source workflow

TIME

PER WEEK	4 hrs
DURATION	2–4 months

PREREQUISITES

C/C++, serial protocols

OUTCOMES

Open-source library · Skill development

COMMERCIAL OUTLOOK

Low alone; medium if RTK-capable.

TECH STACK

C/C++ · Embedded Rust (optional)

ESC Hardware / Firmware

Power electronics · BLDC motor control

Build or flash an Electronic Speed Controller for a brushless motor. Path 1: flash AM32 / BLHeli_32 on an off-the-shelf ESC and tune for a specific motor. Path 2: design your own board, write FOC firmware, prove it on the bench.

WHAT YOU'LL BUILD

- Off-the-shelf ESC flashed and characterised (Path 1), OR
- Custom ESC board in KiCad with FOC firmware (Path 2)
- Bench-test report: motor + propeller load characterisation
- Power-efficiency comparison vs stock firmware
- Schematics and firmware on GitHub

SKILLS YOU'LL GAIN

- BLDC and PMSM motor theory
- Field-oriented control (FOC) math
- Power-electronics basics
- KiCad (Path 2)
- Hard real-time embedded C

TIME

PER WEEK	4 hrs
DURATION	2–4 months

PREREQUISITES

Embedded C, basic power electronics

OUTCOMES

Working ESC · Skill development

COMMERCIAL OUTLOOK

Low individually, but a valuable specialist skill.

TECH STACK

C · AM32 · FOC

GROUP B . SOLO MODULE · PROJECT 11 OF 14

LoRa Telemetry Bridge

Long-range sub-GHz comms

Two ESP32 plus RFM95 boards. Push MAVLink or sensor telemetry over LoRa. Characterise the link across DTU campus. Build a simple repeater that extends range. Output is a measurement dataset plus a small library used by the MANET and multi-bearer teams.

WHAT YOU'LL BUILD

- Bidirectional LoRa link with two ESP32 + RFM95 nodes
- Range dataset: RSSI/SNR at 100 m, 500 m, 1 km, 3 km, 5 km
- Repeater node with collision avoidance
- Python plotting and analysis
- Library packaging for team-project consumption

SKILLS YOU'LL GAIN

- LoRa fundamentals
- ESP32 with Arduino / ESP-IDF
- Antenna basics
- Measurement methodology
- Library API design

TIME

PER WEEK	3–4 hrs
DURATION	2–3 months

PREREQUISITES

Basic C/C++, breadboarding

OUTCOMES

Working prototype · Feeds into MANET / comms projects

COMMERCIAL OUTLOOK

Low alone; reusable building block.

TECH STACK

ESP32 · RadioLib · MAVLink

GROUP B . SOLO MODULE · PROJECT 12 OF 14

Single-Sensor Driver Module

Sensor integration · ROS2 driver

Pick one sensor (LiDAR, thermal, ultrasound array, or stereo camera) and write a really clean ROS2 driver for it. The deliverable isn't just data coming out; it's calibration, time sync, error handling, and a visualisation that proves the driver is good.

WHAT YOU'LL BUILD

- ROS2 driver node for chosen sensor
- Calibration routine specific to the sensor
- Time synchronisation with system clock
- Live RViz visualisation
- README with mounting instructions and limitations
- Integration test with a second sensor

SKILLS YOU'LL GAIN

- ROS2 node architecture
- Sensor calibration techniques
- Time synchronisation (PTP basics)
- RViz visualisation
- Robotics error handling

TIME

PER WEEK	4 hrs
DURATION	2–3 months

PREREQUISITES

Python or C++, basic Linux

OUTCOMES

Reusable driver · Open-source release

COMMERCIAL OUTLOOK

Low alone; reusable across team projects.

TECH STACK

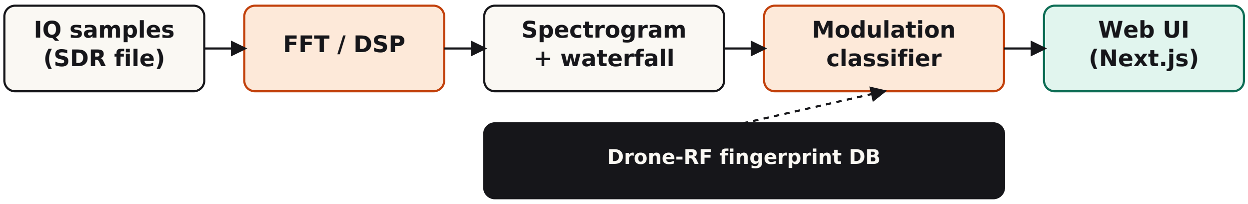
ROS2 · Python / C++

RF Spectrum Analysis Tool

Signal analysis · software

A complete software tool for analysing RF captures from an SDR. Input: IQ-sample dumps or a live SDR feed. Output: spectrograms, modulation classification, drone-RF fingerprinting. Web UI on top. Feeds the anti-drone project directly.

ARCHITECTURE



WHAT YOU'LL BUILD

- IQ-sample reader (RTL-SDR / HackRF / USRP formats)
- Spectrogram + waterfall plot generator
- Modulation classifier (FM, GFSK, FHSS)
- Drone-RF fingerprint library
- Next.js web UI with file upload + live results

SKILLS YOU'LL GAIN

- DSP fundamentals (FFTs, filtering, demodulation)
- GNU Radio companion-app development
- Signal classification
- Full-stack web
- RF capture file formats

TIME

PER WEEK	5 hrs
DURATION	3–5 months

PREREQUISITES

Python, basic DSP

OUTCOMES

Working tool · Research paper potential

COMMERCIAL OUTLOOK

Medium. Anti-drone analytics.

TECH STACK

Python · GNU Radio · NumPy/SciPy · Next.js

Network Analysis Toolkit

Comms diagnostics · packet analysis

A diagnostic toolkit for the lab's comms links. Captures MAVLink / MQTT / UDP traffic, classifies packets by protocol, plots latency distributions (p50 / p99 / p99.9), detects loss patterns. Internal but essential. The multi-bearer and MANET teams depend on it.

ARCHITECTURE

WHAT YOU'LL BUILD

- Packet capture + protocol classification (MAVLink/MQTT/UDP)
- Latency-per-packet measurement with timestamp matching
- Loss-pattern detector (Gilbert-Elliott model fitting)
- Statistical reports: CDFs and percentiles
- Web dashboard for interactive exploration
- Export to PDF for inclusion in research papers

SKILLS YOU'LL GAIN

- Scapy / libpcap analysis
- Wireshark display-filter logic
- Statistical analysis (CDFs, percentiles)
- Frontend data viz (React + d3)
- Research-grade tool design

TIME

PER WEEK	4 hrs
DURATION	2–3 months

PREREQUISITES

Python, basic networking

OUTCOMES

Internal tool · Skill development

COMMERCIAL OUTLOOK

Low alone; reusable across team projects.

TECH STACK

Python · Scapy · Wireshark · React